Enes Birlik

ROBOTICS SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Mechatronics Engineer with hands-on experience in ROS2-based autonomous navigation, motion planning and robotic systems. Experienced in developing ROS2, Nav2, MoveIt2 and mission-planning solutions on real robotic platforms using C++ and Python. Skilled in IoT-based control and communication for industrial systems, user interfaces, hardware integration and Linux-based software development. Practical robotics development experience backed by award-winning TEKNOFEST projects and international competitions including the European Rover Challenge.

CORE COMPETENCIES

Programming Languages: Python, C, C++, C#, MATLAB
Frameworks & Tools: ROS/ROS2, Nav2, MoveIt2, ros2_control, micro-ROS, Docker, Ubuntu/Linux, Git
Hardware & Embedded Systems: Teensy, ESP32, STM32, Raspberry Pi, encoders, servos, I2C/UART, basic electronics
Simulation & Design: Gazebo, MuJoCo, RViz, Blender, SLAM Toolbox, RTAB-Map, SolidWorks
Control & Perception: PID, LQR, Fuzzy Control, sensor fusion (IMU/GPS/LiDAR/Stereo Camera)
Interfaces & Visualization: Qt, LVGL, WPF, Digital Twin / RViz dashboards
Languages: Turkish (Native), English (Upper-Intermediate / B2)
Driving License: Class B, A2
Soft Skills: Problem solving, team leadership, project management

EXPERIENCE

R&D SOFTWARE ENGINEER

Kristal Endüstriyel *September 2026 – Present*

- Contributed to building IoT-based control, communication and monitoring infrastructure by developing software, user interfaces and hardware integration for STM32-based industrial systems.
- Developed touchscreen machine interfaces, PC-based test/simulation tools and data management solutions using LVGL, C/C++ and Python.
- Performed debugging, performance optimization, serial communication and firmware validation on existing systems, testing new features on physical hardware.

SOFTWARE ENGINEER INTERN

STM *July 2025 – August 2025*

- Developed cross-platform C++ applications for robotic systems, ensuring stable hardware communication.
- Designed and tested Qt-based GUI applications, improving usability and reducing interaction latency.
- Debugged serial communication and real-time data handling for embedded device integration into system-level applications.

SOFTWARE ENGINEER INTERN

MARS Autonomous Mobile Robots *July 2024 – August 2024*

- Developed WPF-based interfaces and LiDAR visualization solutions for industrial autonomous mobile robots using C#.
- Gained hands-on experience with industrial AMR platforms and autonomous navigation systems.

TEAM LEADER

KOU ROVER *October 2024 – June 2025*

- Led a 35+ member interdisciplinary engineering team for the ERC international competition.
- Managed full-stack integration of rover subsystems: 6-DOF manipulator, ROS2 navigation stack, vision system and autonomous mission software.
- Developed real-time digital twin visualization using RViz/MoveIt GUI for field debugging and monitoring.
- Coordinated technical documentation, field testing protocols and competition strategy.

SOFTWARE TEAM LEAD

KOU ROVER *September 2023 – October 2024*

- Led the robotics software team implementing a ROS2 navigation stack with Nav2, AMCL localization and SLAM Toolbox.
- Integrated an IMU + GPS + LiDAR + ZED2 stereo camera fusion pipeline.
- Developed computer vision solutions and autonomous mission algorithms.
- Managed team development operations and technical documentation.

TEAM MEMBER
KOU ROVER *October 2021 – September 2023*

- Developed robotic systems including SLAM algorithms (RTAB-Map, Cartographer, SLAM Toolbox), AGV testing and ROS applications.
- Created Gazebo and Blender simulation environments and 3D models across mechanical, electronics and software teams.

PROJECTS

Hand-Tracking Robot Arm Control — MuJoCo + Camera Vision
Personal Project

- Simulated the ALOHA robotic arm in the MuJoCo physics environment for motion planning and control research.
- Implemented real-time hand tracking using camera-based computer vision to drive joint commands.
- Mapped hand pose keypoints to robot arm joint angles, enabling intuitive teleoperation without a physical controller.

Modular 6-DOF Robotic Arm — ROS2 + MoveIt2 (ERC Mission)
KOU ROVER — ERC Competition

- Built a full Gazebo simulation environment including a hardware interface layer with ros2_control.
- Implemented inverse kinematics and motion planning with MoveIt2; teleoperation via a joystick ROS2 command layer.
- Implemented camera-based object detection and filtering for the mission.
- Developed real-time digital twin visualization (RViz / MoveIt GUI) for field debugging.

Autonomous Mobile Robot Platform — ROS1/ROS2 Navigation Stack
TEKNOFEST + KOU ROVER Missions

- Full-stack autonomous rover: sensor fusion (IMU + GPS + LiDAR + ZED2), SLAM, navigation, mission execution.
- Mapping with Nav2 + SLAM Toolbox, RTAB-Map and RPLidar scan matching.
- Custom Gazebo + Blender simulation environments; lane detection and a decision state machine.
- Remote operation system with a custom user interface; field testing and electronics assembly.

ROS2–MATLAB Integrated Inverted Pendulum Control System
Personal Project

- Designed a hybrid control system for inverted pendulum stabilization with a ROS2 hardware interface and actuator communication.
- Implemented and compared PID, LQR and Fuzzy controllers in MATLAB and Python.
- Demonstrated real-time closed-loop ROS2 and MATLAB interoperability.

MILITARY SERVICE

Turkish Armed Forces | *Completed*

EDUCATION

B.Sc. Mechatronics Engineering
Kocaeli University *2020 – 2025* | *GPA: 3.04*

High School Diploma
Aldemir Atilla Konuk Anatolian High School *2016 – 2020* | *Graduation Grade: 92.48*

AWARDS & RESEARCH

TEKNOFEST — Digital Technologies in Industry Competition
1st Place (2024, 2025) • 3rd Place (2022) • 6th Place (2023)

Humanoid Robot Development — TÜBİTAK 2209-A
Project Team Member

6-Axis Robotic Arm Development — TÜBİTAK 2209-A
Project Team Member

European Rover Challenge (ERC)
International Participant — KOU ROVER Team